Lecture 7

Angular momentum: the multiplets of spin j

Objectives. *Rotations about different axes do not commute. When they act on a system by unitaries that do not commute either, Lecture 6 shows that a Hamiltonian unchanged by them has degenerate levels. This lecture finds which degeneracies can occur. An angular momentum is a triple of observables with the commutation relations of the spin of Lecture 1. We prove that every angular momentum on $\mathbb{C}^n$ splits into multiplets of spin j, of dimension $2j+1$, and that a multiplet is fixed by j alone. The three-level system of Lecture 4 turns out to be a multiplet of spin 1. If the Hamiltonian commutes with time reversal, every level of a system of half-integer spin is evenly degenerate.*

By the end of this lecture you will be able to:

- Find the raising and lowering matrices of an angular momentum, and the norms of the vectors they produce.
- Find the multiplets of an angular momentum, with their multiplicities, from the eigenvalues of J_z.
- Write the matrices of spin $\frac{1}{2}$ and spin 1 in the standard basis.
- Find the degeneracies that rotations, or time reversal, force on the levels of a Hamiltonian.

7.1 Angular momentum

The spin of Lecture 1 is the triple $S_a = \frac{\hbar}{2}\sigma_a$. By the Pauli rule, $\sigma_x\sigma_y - \sigma_y\sigma_x = 2\mathrm{i}\sigma_z$, so $S_xS_y - S_yS_x = \mathrm{i}\hbar S_z$, and the same holds for the cyclic permutations $(x,y,z) \to (y,z,x) \to (z,x,y)$. These relations come from rotations. By Theorem 1.12, the unitary $e^{-\mathrm{i}\theta\,\mathbf{n}\cdot\boldsymbol{\sigma}/2} = e^{-\mathrm{i}\theta\,\mathbf{n}\cdot\mathbf{S}/\hbar}$ turns the Bloch sphere by the angle θ about $\mathbf{n}$. Rotations about different axes do not commute, and neither do their generators. For a system larger than a qubit, the observables that generate its rotations in the same way are its angular momentum, and they must obey the same relations. We take the relations as the definition. Exercise Sheet 7 shows that they make $e^{-\mathrm{i}\theta J_z/\hbar}$ turn the pair (J_x, J_y) through the angle θ.

Definition 7.1 (Angular momentum)**.** An *angular momentum* on $\mathbb{C}^n$ is a triple of self-adjoint matrices J_x, J_y, J_z with

$$\big[J_x, J_y\big] = \mathrm{i}\hbar J_z, \qquad \big[J_y, J_z\big] = \mathrm{i}\hbar J_x, \qquad [J_z, J_x] = \mathrm{i}\hbar J_y. \tag{7.1}$$

The spin of Lecture 1 is an angular momentum on $\mathbb{C}^2$. So is the zero triple on any $\mathbb{C}^n$. Consider atoms whose magnetic moment is $\gamma\mathbf{J}$, for an angular momentum $\mathbf{J}$ and a number

$\gamma \neq 0$. A Stern–Gerlach magnet along z reads J_z on them. Silver gave two beams. The theorem of this lecture says which other numbers of beams are possible.

7.2 Raising and lowering

For an angular momentum $\mathbf{J}$, define

$$J^2 = J_x^2 + J_y^2 + J_z^2, \qquad J_\pm = J_x \pm \mathrm{i}J_y. \tag{7.2}$$

The matrix J^2 is self-adjoint, and $J_\pm^\dagger = J_\mp$. The names J_+ and J_- come from the next lemma. They raise and lower the reading of J_z by $\hbar$.

Lemma 7.2 (Raising and lowering). *Let $\mathbf{J}$ be an angular momentum on $\mathbb{C}^n$.*

1. *J^2 commutes with J_x, J_y and J_z.*
2. *$[J_z, J_\pm] = \pm\hbar J_\pm$, and*
$$J_\mp J_\pm = J^2 - J_z^2 \mp \hbar J_z. \tag{7.3}$$
3. *Let v be a unit vector with $J^2 v = \hbar^2 \lambda v$ and $J_z v = \hbar m v$. Then $J_\pm v$ satisfies $J^2(J_\pm v) = \hbar^2 \lambda J_\pm v$ and $J_z(J_\pm v) = \hbar(m \pm 1)J_\pm v$, and*
$$\|J_\pm v\|^2 = \hbar^2\big(\lambda - m(m \pm 1)\big). \tag{7.4}$$

Proof.

1. For any matrices A and B, $\left[A^2, B\right] = A\,[A, B] + [A, B]\,A$. So
$$\left[J_x^2, J_z\right] = -\mathrm{i}\hbar(J_x J_y + J_y J_x), \qquad \left[J_y^2, J_z\right] = \mathrm{i}\hbar(J_y J_x + J_x J_y),$$
using $[J_x, J_z] = -\mathrm{i}\hbar J_y$ and $\left[J_y, J_z\right] = \mathrm{i}\hbar J_x$. The two cancel, and $\left[J_z^2, J_z\right] = 0$, so $\left[J^2, J_z\right] = 0$. The cyclic permutation of the relations gives the same for J_x and J_y.
2. First, $\left[J_z, J_x \pm \mathrm{i}J_y\right] = \mathrm{i}\hbar J_y \pm \mathrm{i}(-\mathrm{i}\hbar J_x) = \pm\hbar(J_x \pm \mathrm{i}J_y)$. Next,
$$J_\mp J_\pm = (J_x \mp \mathrm{i}J_y)(J_x \pm \mathrm{i}J_y) = J_x^2 + J_y^2 \pm \mathrm{i}\left[J_x, J_y\right] = J^2 - J_z^2 \mp \hbar J_z.$$
3. By the first part, $J^2 J_\pm v = J_\pm J^2 v = \hbar^2 \lambda J_\pm v$. By the second, $J_z J_\pm v = J_\pm J_z v \pm \hbar J_\pm v = \hbar(m \pm 1)J_\pm v$. Finally, $\|J_\pm v\|^2 = \langle J_\pm v, J_\pm v\rangle = \langle v, J_\mp J_\pm v\rangle$, and (7.3) gives $\hbar^2(\lambda - m^2 \mp m)\,\|v\|^2$. ∎

Since a norm is never negative, (7.4) gives $\lambda \geq m(m+1)$ and $\lambda \geq m(m-1)$ whenever $\hbar m$ and $\hbar^2\lambda$ are joint eigenvalues of J_z and J^2. Therefore the readings of J_z at a fixed λ are bounded. This is why repeated raising or lowering must stop.

7.3 The multiplets

Theorem 7.3 (The multiplets of spin j). *Let $\mathbf{J}$ be an angular momentum on $\mathbb{C}^n$, $n \geq 1$. Then $\mathbb{C}^n$ is an orthogonal sum of subspaces $V_1, \ldots, V_r$, each invariant under J_x, J_y and J_z, with the following property. Each V_i has a number j_i, with $2j_i$ a non-negative integer, and an orthonormal basis $|j_i, m\rangle$, $m = j_i,\ j_i - 1,\ \ldots,\ -j_i$, in which*

$$J_z\,|j, m\rangle = \hbar m\,|j, m\rangle\,, \qquad J_\pm\,|j, m\rangle = \hbar\sqrt{(j \mp m)(j \pm m + 1)}\,|j, m \pm 1\rangle\,, \tag{7.5}$$

with $j = j_i$ and $|j, \pm(j+1)\rangle = 0$. On V_i, $J^2 = \hbar^2 j_i(j_i+1)\mathbb{1}$. For each j, the number of the V_i with $j_i = j$ is $d_j - d_{j+1}$, where d_m is the dimension of the eigenspace of J_z for the eigenvalue

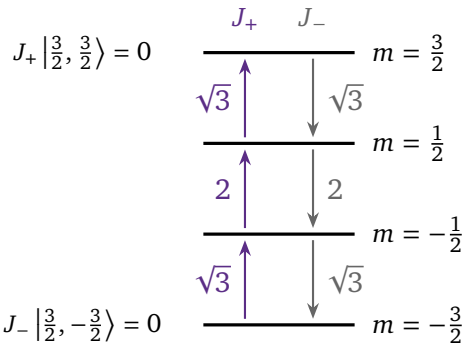


Figure 7.1: The multiplet of spin $\frac{3}{2}$. Each rung is a vector $\left|\frac{3}{2}, m\right\rangle$ of the standard basis. The matrix J_+ moves up a rung and J_- down, with the coefficients of (7.5) in units of $\hbar$. The ladder stops at both ends.

> $\hbar m$.

Each V_i is a *multiplet of spin* j_i, of dimension $2j_i + 1$. The basis $|j, m\rangle$ is its *standard basis*. Its phases follow a convention due to Condon and Shortley, which makes the coefficients in (7.5) positive. By (7.5), in the standard bases of two multiplets of the same spin j, the matrices of the three J_a are the same. So the unitary that takes one standard basis to the other carries each J_a on the first multiplet to J_a on the second. A multiplet is fixed by j alone, up to this unitary equivalence. Figure 7.1 shows the multiplet of spin $\frac{3}{2}$.

Proof. We build one multiplet, show that what remains is again an angular momentum, and repeat. Throughout, Lemma 7.2 is used for the present angular momentum.

Step 1: a top vector. The matrices J^2 and J_z are self-adjoint and commute, so they have a common eigenvector (*Linear Algebra*, Prop. 6.8). Dividing it by its norm gives a common unit eigenvector v. By Lemma 7.2, each nonzero vector among v, J_+v, J_+^2v, … is again a common eigenvector, and their eigenvalues for J_z go up by $\hbar$ each time. These eigenvalues are distinct, and J_z has at most n of them, so $J_+^k v = 0$ for some $k \geq 1$. Let u be the last nonzero vector of the list, divided by its norm. Then u is a common unit eigenvector with $J_+u = 0$. Write $J_zu = \hbar ju$ and $J^2u = \hbar^2\lambda u$. By (7.3),

$$0 = J_-J_+u = (J^2 - J_z^2 - \hbar J_z)u = \hbar^2\big(\lambda - j(j+1)\big)u,$$

so $\lambda = j(j+1)$.

Step 2: lowering. Put $u_j = u$. Suppose that u_m is a common unit eigenvector, with $J^2u_m = \hbar^2 j(j+1)u_m$ and $J_zu_m = \hbar m u_m$, where $j - m$ is a non-negative integer. By (7.4),

$$\|J_-u_m\|^2 = \hbar^2\big(j(j+1) - m(m-1)\big) = \hbar^2(j+m)(j-m+1).$$

Here $j - m + 1 \geq 1$. So the norm is 0 exactly when $m = -j$, and if it is not 0, then $j + m > 0$. In that case put

$$c_m = \sqrt{(j+m)(j-m+1)} > 0, \qquad u_{m-1} = \frac{J_-u_m}{\hbar c_m}.$$

By Lemma 7.2, u_{m-1} is a common unit eigenvector with the J_z-eigenvalue $\hbar(m-1)$, and the step can be repeated. The vectors u_j, u_{j-1}, … have distinct eigenvalues for J_z, so the list must stop, and it stops only at a vector u_{-j} with $J_-u_{-j} = 0$. The list went from $m = j$ to $m = -j$ in steps of 1, so $2j$ is a non-negative integer, and there are $2j + 1$ vectors. They are orthonormal, since they are unit eigenvectors of the self-adjoint J_z with distinct eigenvalues.

Step 3: the multiplet. By construction, $J_- u_m = \hbar c_m u_{m-1}$ for $m > -j$, and $J_- u_{-j} = 0$. Apply J_+ and use (7.3) with the lower signs:

$$\hbar c_m \, J_+ u_{m-1} = J_+ J_- u_m = (J^2 - J_z^2 + \hbar J_z) u_m = \hbar^2 \big(j(j+1) - m^2 + m\big) u_m = \hbar^2 c_m^2 \, u_m .$$

Dividing by $\hbar c_m > 0$ gives $J_+ u_{m-1} = \hbar c_m u_m$, and $J_+ u_j = 0$. With $|j,m\rangle = u_m$ these are the formulas (7.5), since $c_m = \sqrt{(j-(m-1))(j+(m-1)+1)}$. So the span V_1 of the u_m is invariant under J_z, J_+ and J_-, hence under $J_x = \frac{1}{2}(J_+ + J_-)$ and $J_y = \frac{1}{2\mathrm{i}}(J_+ - J_-)$. Every u_m has $J^2 u_m = \hbar^2 j(j+1) u_m$, so $J^2 = \hbar^2 j(j+1)\mathbb{1}$ on V_1.

Step 4: the rest. Let W be the orthogonal complement of V_1. For $w \in W$ and $v \in V_1$, $\langle v, J_a w\rangle = \langle J_a v, w\rangle = 0$, since $J_a v$ lies in V_1. So W is invariant under every J_a. In an orthonormal basis of W, the restrictions of the J_a to W are self-adjoint matrices, and they obey (7.1), since products of restrictions are restrictions of products. So they form an angular momentum on W. If $W \neq 0$, Steps 1–3 applied to W give a second multiplet $V_2 \subset W$, and so on. The dimension left drops at each step, so after finitely many steps $\mathbb{C}^n = V_1 \oplus \cdots \oplus V_r$.

Step 5: counting. By (7.5), the eigenspace of J_z for $\hbar m$ is spanned by the vectors $|j_i, m\rangle$ of those multiplets with $j_i \geq |m|$ and $j_i - m$ an integer, one vector from each. So d_j counts the multiplets with $j_i \geq j$ and $j_i - j$ an integer, and d_{j+1} those with $j_i \geq j+1$ and the same integrality. The difference $d_j - d_{j+1}$ counts the multiplets with $j_i = j$. ∎

The count in Step 5 shows that the spins j_i that occur, with their multiplicities, do not depend on the choices made in the proof. Hence J^2 and J_z together form a complete set (Definition 5.4) exactly when no spin occurs twice. Two multiplets of the same j need a further label to tell their vectors $|j,m\rangle$ apart.

7.4 Spin $\frac{1}{2}$ and spin 1

Example 7.4 (Spin $\frac{1}{2}$ and spin 1)**.**

1. *Spin* $\frac{1}{2}$. For $j = \frac{1}{2}$, the formulas (7.5) give $J_z = \frac{\hbar}{2}\sigma_z$ and $J_+ = \hbar\,|{\uparrow}\rangle\langle{\downarrow}|$ in the basis $|{\uparrow}\rangle = \big|\frac{1}{2}, \frac{1}{2}\big\rangle$, $|{\downarrow}\rangle = \big|\frac{1}{2}, -\frac{1}{2}\big\rangle$, so $J_a = \frac{\hbar}{2}\sigma_a$, the spin of Lecture 1. On $\mathbb{C}^2$ the dimensions $2j+1$ of the multiplets add up to 2, so every angular momentum on $\mathbb{C}^2$ is either 0 or, up to a unitary, the spin of Lecture 1.
2. *Spin* 1. For $j = 1$, in the basis $|1,1\rangle$, $|1,0\rangle$, $|1,-1\rangle$, the coefficients are $\hbar\sqrt{2}$, and
$$J_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \qquad J_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad J_y = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -\mathrm{i} & 0 \\ \mathrm{i} & 0 & -\mathrm{i} \\ 0 & \mathrm{i} & 0 \end{pmatrix}.$$
3. *The three-level system is spin* 1. Identify $e_+ = |1,1\rangle$, $e_0 = |1,0\rangle$ and $e_- = |1,-1\rangle$. Then the observables of Lecture 4 are
$$Z = \frac{J_z}{\hbar}, \qquad H = \frac{\varepsilon}{\hbar^2} J_z^2, \qquad X = \frac{J_x^2 - J_y^2}{\hbar^2}.$$
The rotations $V_\alpha = e^{-\mathrm{i}\alpha Z} = e^{-\mathrm{i}\alpha J_z/\hbar}$ of Lecture 6 are generated by J_z, and the mirror is $M = -e^{-\mathrm{i}\pi J_x/\hbar}$, a rotation by π about x up to a sign (Exercise Sheet 7). So H commutes with J_z but not with J_x.

A spin j with $2j$ odd is called *half-integer*, and one with j an integer is called *integer*. In a multiplet of half-integer spin, $e^{-2\pi\mathrm{i}J_z/\hbar}$ multiplies every $|j,m\rangle$ by $e^{-2\pi\mathrm{i}m} = -1$. A rotation by 2π gives $-\mathbb{1}$, as for the qubit in Lecture 1. So the states of spin $\frac{1}{2}$ do not turn like vectors in space, which a rotation by 2π leaves unchanged. For integer j, the same rotation gives $\mathbb{1}$.

7.5 Rotations and degeneracy

Corollary 7.5 (Degeneracies forced by rotations)**.** *Let H commute with J_x, J_y and J_z. Then every eigenspace of H is an orthogonal sum of multiplets, and the degeneracy of every level of H is a sum of numbers $2j+1$. If $\mathbb{C}^n$ is a single multiplet, H is a multiple of $\mathbb{1}$.*

Proof. An eigenspace of H is invariant under every J_a, by the proof of Lemma 4.3, and the restrictions form an angular momentum there, as in Step 4. Theorem 7.3 applied to it gives the first claim. On a single multiplet, H commutes with J_z, whose eigenvalues $\hbar m$ are simple, so $H|j,m\rangle = h_m|j,m\rangle$ for numbers h_m. Then $HJ_-|j,m\rangle = J_-H|j,m\rangle$ gives $h_{m-1}\hbar c_m = h_m \hbar c_m$, and $c_m > 0$, so all the h_m are equal. ∎

Without the hypothesis that H commutes with all three J_a, the corollary forces nothing. The Hamiltonian $H = (\varepsilon/\hbar^2)J_z^2$ of the three-level system acts on a single multiplet of spin 1, yet it has two levels, 0 and ε. Of the three J_a, it commutes with J_z only. The degeneracy of its level ε comes from the rotations about the z axis together with the mirror (Lecture 6), not from the full set of rotations.

When every rotation is a symmetry, as for an atom in no field, each level contains a whole multiplet, so a level that contains a multiplet of spin j is at least $(2j+1)$-fold degenerate. A field along z breaks the rotations about x and y, and it can split the level.

The physics

The electron, the proton and the neutron each have spin $\frac{1}{2}$. The deuteron, the nucleus of heavy hydrogen, and the nucleus of nitrogen-14 have spin 1, and so has the pair of unpaired electrons in an NV^- centre, the three-level system of Lecture 4. In Stern and Gerlach's experiment the electronic angular momentum of a silver atom is that of one electron, spin $\frac{1}{2}$, and the beam splits in two. An ideal magnet for a single multiplet of spin j, whose magnetic moment is $\gamma\mathbf{J}$ with $\gamma \neq 0$, separates up to $2j+1$ beams, one for each reading of J_z, and an unpolarised beam fills all of them. Chromium ions in ruby, Cr^{3+}, have spin $\frac{3}{2}$. The crystal splits their lowest multiplet into two Kramers pairs (§ 7.6) about 11.5 GHz apart. These levels, split further by an applied field, are used in ruby masers.

7.6 Time reversal

Run a motion backwards and every spin turns round. This reversal is not a unitary matrix. A map Θ of $\mathbb{C}^n$ onto itself is *antiunitary* if it is antilinear and reverses the order in inner products, that is, if for all vectors ϕ, ψ and all numbers α, β

$$\Theta(\alpha\phi + \beta\psi) = \bar{\alpha}\,\Theta\phi + \bar{\beta}\,\Theta\psi \quad \text{and} \quad \langle \Theta\phi, \Theta\psi\rangle = \langle \psi, \phi\rangle\,.$$

Let C conjugate the coordinates in an orthonormal basis. The product $U = \Theta C$ of two antilinear maps is linear, and it keeps inner products, so it is unitary. Since $C^2 = \mathbb{1}$, we have $\Theta = UC$. Since $CU = \bar{U}C$, it follows that $\Theta^2 = UCUC = U\bar{U}$, where $\bar{U}$ has the conjugate entries. If $\Theta H = H\Theta$, then Θ maps each eigenspace of H into itself. Indeed, if $H\psi = \lambda\psi$, then $H\Theta\psi = \Theta H\psi = \Theta(\lambda\psi) = \lambda\Theta\psi$, since λ is real.

Definition 7.6 (Time reversal)**.** On a sum of multiplets with their standard bases (Theo-

rem 7.3), *time reversal* is the map Θ that acts on each multiplet by

$$\Theta\Big(\sum_m c_m \,|j,m\rangle\Big) = \sum_m \bar{c}_m \,(-1)^{j-m}\,|j,-m\rangle\,.$$

Lemma 7.7 (Time reversal reverses angular momentum). *Time reversal Θ is antiunitary, $\Theta J_a \Theta^{-1} = -J_a$ for $a = x, y, z$, and $\Theta^2 = (-1)^{2j}\mathbb{1}$ on each multiplet of spin j.*

Proof. The map Θ conjugates the coordinates and then permutes the standard basis vectors with signs, so it is antiunitary. It takes $|j,m\rangle$ to $\pm\,|j,-m\rangle$, so $\Theta J_z \Theta^{-1} = -J_z$. Since $\Theta^{-1}\,|j,m\rangle = (-1)^{j+m}\,|j,-m\rangle$, and the coefficients of (7.5) are real,

$$\Theta J_+ \Theta^{-1}\,|j,m\rangle = (-1)^{j+m}(-1)^{j+m-1}\hbar\sqrt{(j+m)(j-m+1)}\,|j,m-1\rangle = -J_-\,|j,m\rangle\,,$$

since $j+m$ is an integer. Likewise, $\Theta J_- \Theta^{-1} = -J_+$. Then $\Theta J_x \Theta^{-1} = \frac{1}{2}\Theta(J_+ + J_-)\Theta^{-1} = -J_x$. Since Θ conjugates the number $\frac{1}{2\mathrm{i}}$,

$$\Theta J_y \Theta^{-1} = -\frac{1}{2\mathrm{i}}\,\Theta(J_+ - J_-)\Theta^{-1} = -\frac{1}{2\mathrm{i}}(-J_- + J_+) = -J_y.$$

Finally $\Theta^2\,|j,m\rangle = (-1)^{j-m}(-1)^{j+m}\,|j,m\rangle = (-1)^{2j}\,|j,m\rangle$. ∎

In the three-level system, $H + \eta X = (\varepsilon/\hbar^2)J_z^2 + (\eta/\hbar^2)(J_x^2 - J_y^2)$ commutes with Θ, since $\Theta J_a^2 \Theta^{-1} = (-J_a)^2 = J_a^2$ for each a and the numbers ε and η are real. A field term bZ does not, since $\Theta Z \Theta^{-1} = -Z$.

Theorem 7.8 (Kramers' degeneracy). *Let Θ be antiunitary with $\Theta^2 = -\mathbb{1}$, and let Θ commute with a self-adjoint H. Then every eigenspace of H has even dimension.*

Proof. By antiunitarity,

$$\langle \psi, \Theta\psi\rangle = \big\langle \Theta^2\psi, \Theta\psi\big\rangle = -\,\langle \psi, \Theta\psi\rangle\,,$$

so $\Theta\psi \perp \psi$. In an eigenspace $\mathcal{V}$ take a unit vector ψ_1. Then $\Theta\psi_1 \in \mathcal{V}$ is a unit vector orthogonal to it. If $\phi \in \mathcal{V}$ is orthogonal to both, then so is $\Theta\phi$. Indeed, $\langle \Theta\phi, \Theta\psi_1\rangle = \langle \psi_1, \phi\rangle = 0$, and, since $\psi_1 = \Theta(-\Theta\psi_1)$, $\langle \Theta\phi, \psi_1\rangle = -\,\langle \Theta\psi_1, \phi\rangle = 0$. So Θ maps the part of $\mathcal{V}$ orthogonal to ψ_1 and $\Theta\psi_1$ into itself. Repeat the argument there, two dimensions at a time. ∎

On a sum of multiplets that all have half-integer spin, $\Theta^2 = -\mathbb{1}$, so a Hamiltonian that commutes with Θ has every level evenly degenerate. On a single spin $\frac{1}{2}$, such a Hamiltonian is therefore a multiple of $\mathbb{1}$. For integer spin $\Theta^2 = \mathbb{1}$, and Theorem 7.8 does not apply. For example, the strain ηX commutes with Θ and splits the level of e_+ and e_- (Example 6.6). A magnetic field breaks time reversal and can split a Kramers pair, as in Lecture 1. Since it is antilinear, Θ is not a symmetry in the sense of Lecture 6 and has no generator.

Angular momentum on $\mathbb{C}^n$

Definition: self-adjoint J_x, J_y, J_z with $[J_x, J_y] = \mathrm{i}\hbar J_z$ and its cyclic permutations. *Structure:* $\mathbb{C}^n$ is an orthogonal sum of multiplets of spin j, $2j \in \{0, 1, 2, \dots\}$, each of dimension $2j+1$ with the standard basis $|j,m\rangle$ and $J^2 = \hbar^2 j(j+1)\mathbb{1}$ on it. The multiplicity of j is $d_j - d_{j+1}$ (Theorem 7.3). *Degeneracy:* a Hamiltonian commuting with all J_a has levels whose degeneracies are sums of $2j+1$ (Corollary 7.5).

Summary.

- *Raising and lowering*: $J_\pm$ move the reading of J_z by $\pm\hbar$ inside an eigenspace of J^2, and their norms are fixed by (7.4) (Lemma 7.2).
- *The multiplets of spin j*: every angular momentum on $\mathbb{C}^n$ is an orthogonal sum of multiplets, each fixed by its spin j up to a unitary, and the multiplicities are $d_j - d_{j+1}$ (Theorem 7.3).
- *Degeneracies forced by rotations*: sums of $2j+1$ (Corollary 7.5).
- *Kramers' degeneracy*: if Θ is antiunitary, $\Theta^2 = -\mathbb{1}$ and $\Theta H = H\Theta$, every level of H is evenly degenerate (Theorem 7.8).

Lecture 8 turns to a particle that hops on a ring, where translations, not rotations, fix the levels.

References

Sakurai and Napolitano, *Modern Quantum Mechanics*, §3.5, pp. 180–186. Cohen-Tannoudji, Diu and Laloë, *Quantum Mechanics*, vol. 1, Ch. VI, §C, pp. 671–684. Binney and Skinner, *The Physics of Quantum Mechanics*, p. 182. Sakurai and Napolitano, §4.4, pp. 270–284. Landau and Lifshitz, *Quantum Mechanics: Non-relativistic Theory*, §60, pp. 223–226. Kramers, *Proceedings Amsterdam* 33, 959 (1930).